

LOGRONO AGONCILLO AP, Spain

WMO: 080840

Lat: 42.4522N

Lon: 2.3311W

Elev: 363

StdP: 97.04

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.1	-0.8	-6.4	2.3	2.3	-4.4	2.7	2.8	10.1	8.2	8.7	8.0	0.8	270	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.8	35.0	20.2	33.1	19.8	31.2	19.4	21.8	31.0	21.0	29.9	20.2	28.6	3.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	14.4	24.9	18.1	13.6	24.0	17.2	12.9	23.4	65.3	31.1	62.1	30.0	59.3	28.8	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	7.4	6.3	DB	-4.8	37.9	2.0	1.5	-6.2	39.0	-7.3	39.9	-8.4	40.7	-9.9	41.8	
			WB	-5.2	23.3	2.0	1.5	-6.7	24.3	-7.8	25.2	-8.9	26.0	-10.4	27.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.3	6.6	7.4	10.4	12.6	16.3	20.8	23.1	23.0	19.5	15.3	9.8	6.6
	DBStd	6.76	3.02	2.94	3.11	3.06	3.42	3.65	3.06	2.87	2.83	3.21	3.08	3.15
	HDD10.0	388	111	81	33	10	1	0	0	0	0	2	41	110
	HDD18.3	1932	363	306	245	172	82	17	3	2	19	104	257	363
	CDD10.0	1964	6	8	46	89	195	323	406	402	285	165	33	5
	CDD18.3	467	0	0	0	1	18	91	151	144	54	8	0	0
	CDH23.3	4713	0	0	8	42	249	978	1494	1387	482	73	1	0
	CDH26.7	2017	0	0	1	5	68	435	708	640	152	8	0	0
Wind	WSAvg	2.5	2.7	2.9	2.9	2.8	2.6	2.5	2.4	2.3	2.1	2.1	2.5	2.5
Precipitation	PrecAvg	660	63	56	64	71	65	51	33	22	44	62	75	55
	PrecMax	873	116	118	135	161	212	130	84	70	99	147	151	133
	PrecMin	531	20	20	21	39	10	5	2	1	22	22	34	14
	PrecStd	94	26	27	28	25	36	31	21	16	18	28	26	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.9	18.8	24.1	27.1	31.8	36.5	36.9	37.2	33.4	27.8	20.7	16.3
		MCWB	11.6	11.3	12.9	15.6	18.0	19.8	20.7	20.3	19.4	16.8	13.8	11.2
	2%	DB	14.5	16.4	21.1	24.2	28.5	33.7	34.9	34.7	30.4	25.3	18.0	14.0
		MCWB	10.3	10.5	12.4	14.5	17.1	19.4	20.4	20.0	18.6	16.4	13.0	10.2
	5%	DB	13.0	14.7	19.0	21.8	26.1	31.4	33.0	32.7	28.2	23.1	16.1	12.7
		MCWB	9.4	9.7	11.9	13.5	16.3	19.0	20.2	19.8	17.9	15.8	12.0	9.5
	10%	DB	11.5	12.9	16.9	19.4	23.9	29.0	30.9	30.4	26.1	21.1	14.6	11.2
		MCWB	8.5	8.7	10.8	12.6	15.4	18.5	19.6	19.4	17.4	15.0	11.1	8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	12.5	14.7	16.4	19.7	21.9	23.2	22.9	21.2	18.6	15.2	12.0
		MCDB	15.7	16.6	21.1	25.0	29.0	31.3	33.3	32.2	28.6	24.7	18.9	14.8
	2%	WB	10.9	11.1	13.2	15.2	18.1	20.7	21.9	21.7	20.0	17.6	13.7	10.9
		MCDB	13.6	15.6	19.2	22.5	26.3	30.3	31.4	30.7	27.3	22.8	16.9	13.5
	5%	WB	9.8	10.1	12.3	14.1	17.0	19.9	21.0	20.8	19.0	16.7	12.7	10.0
		MCDB	12.5	13.9	17.8	20.4	24.5	28.9	30.1	29.5	25.8	21.2	15.3	12.0
	10%	WB	8.8	9.1	11.2	13.0	16.0	19.0	20.1	20.0	18.1	15.9	11.7	9.1
		MCDB	11.1	12.1	16.1	18.6	22.5	27.5	28.5	28.1	24.3	19.7	13.9	11.0
Mean Daily Temperature Range	5% DB	MDBR	7.5	9.1	11.1	11.4	12.1	13.2	13.8	13.8	12.4	10.8	7.9	7.3
		MCDBR	10.2	12.7	15.2	16.1	16.7	17.4	17.5	17.6	16.0	14.1	10.4	9.3
		MCWBR	6.5	7.4	7.8	7.7	7.1	6.3	5.8	5.9	6.1	6.6	6.0	6.0
	5% WB	MCDBR	9.0	11.4	13.7	14.3	14.9	15.7	15.6	15.1	13.8	11.6	9.1	8.3
		MCWBR	6.1	7.0	7.4	7.3	6.9	6.3	5.8	5.9	5.8	5.9	5.7	5.8
Clear-Sky Solar Irradiance	taub	0.312	0.330	0.371	0.394	0.408	0.417	0.405	0.408	0.386	0.358	0.328	0.309	
	taud	2.475	2.414	2.303	2.253	2.246	2.256	2.290	2.285	2.345	2.423	2.468	2.484	
	Ebn at Noon	832	871	869	873	867	857	864	850	843	825	803	801	
	Edh at Noon	74	92	116	131	135	135	129	125	110	90	74	68	
All-Sky Solar Radiation	RadAvg	1.65	2.46	3.69	4.73	5.66	6.53	6.82	5.93	4.57	3.02	1.84	1.49	
	RadStd	0.17	0.31	0.47	0.48	0.45	0.38	0.32	0.28	0.20	0.27	0.18	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	-0.63	-0.84	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page